# Problem Set

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Name: ______________________________

1. (10 points) **Principal-Agent Problem with Moral Hazard**

   A principal hires an agent to complete a project. The agent exerts effort $e$. Effort costs the agent $\frac{e^2}{2}$. Both, principal and agent, are risk neutral.

   The project generates revenue $x = 10e$ for the principal. The principal offers a contract $w = a + bx$ where $a$ is the base salary and $b \leq 1$ is the bonus rate.

   The agent's reservation utility is $\underline{u} = 2$.

   (a) (5 points) If effort is observable (first-best), what effort level maximizes total surplus? What contract would implement this?

   (b) (5 points) If effort is not observable (second-best), *i.e.* $x = 10e + u$ with $u \overset{iid}{\sim} (0, \sigma_u^2)$, what effort will the agent choose if offered $b = 0.2$? Is this optimal for the principal?

2. (10 points) **Bayesian Games and Incomplete Information**

   Two countries are negotiating a trade agreement. Country A knows its type (high-value $\theta_H = 10$ or low-value $\theta_L = 4$), but Country B only knows that each type is equally likely ($p = 0.5$).

   Country A can propose either a comprehensive agreement (C) or a limited agreement (L). Country B then accepts (A) or rejects (R).

   Payoffs:

   - If type $\theta_H$ proposes C and B accepts: (8, 6)
   - If type $\theta_H$ proposes C and B rejects: (0, 0)
   - If type $\theta_H$ proposes L and B accepts: (5, 4)
   - If type $\theta_H$ proposes L and B rejects: (1, 0)
   - If type $\theta_L$ proposes C and B accepts: (3, 6)
   - If type $\theta_L$ proposes C and B rejects: (0, 0)
   - If type $\theta_L$ proposes L and B accepts: (4, 4)
   - If type $\theta_L$ proposes L and B rejects: (1, 0)

   (a) (5 points) Is there a separating Perfect Bayesian Equilibrium where type $\theta_H$ plays C and type $\theta_L$ plays L? What would B's best response be?

   (b) (5 points) Is there a pooling Perfect Bayesian Equilibrium where both types play C? What would B's best response be?

3. (10 points) Consider the Dynamic Inconsistency of the Fiscal Policy game. Devise and write down the **extensive form** of the game. Think on discrete space (high tax $\tau^h$, low tax $\tau^l$, high investment $I_h$, low investment $I_l$) etc. You are free to choose how many stages the game has, the payoffs, etc. The idea of this question is to see if you can think on a situation/phenomena and model it using what we covered in this course.